Component 3 Data Strategy (FishGuard AI)

Objective: Develop a machine learning model to forecast tuna availability/catch using fisheries records and environmental indicators.

STEP 1 - DATA SOURCE CLASSIFICATION

During the research, multiple fisheries-related data sources were identified. They are classified into two categories:

Category A - Core Modeling Data

These datasets directly contribute to model training.

Fisheries Logbook Data (Primary Fisheries Dataset)

Contains information such as:

- Species Name
- Weight
- Quantity
- Date
- Vessel ID
- Gear Type
- Fishing Location
- Ports
- Coordinates

Reason for selection:

- Provides detailed fishing activity records.
- Contains spatial and temporal information.
- Allows integration with environmental datasets.
- Supports tuna-specific filtering.
- Suitable for machine learning.

Status: Main fisheries dataset.

Category B - Reference & Supporting Data

Ministry of Fisheries Statistics

Sources: Monthly Fish Production, Annual Fish Production, Fisheries Statistical Bulletins, Fleet Statistics, Infrastructure Statistics

Purpose: Background information, Trend analysis, Research discussion, Result validation, Viva evidence

Not intended as a primary machine learning dataset.

Reason: Data is aggregated. Limited species-level information. Limited spatial information. Cannot directly link catches to environmental conditions.

Status: Supporting dataset.

Export / Import Statistics

Contains: Tuna exports, Export quantities, Export values, Destination countries

Purpose: Industry context, Market discussion, Future work, Economic interpretation

Not directly used for forecasting tuna availability.

Status: Reference only.

STEP 2 - DATA STORAGE STRATEGY

GitHub Repository

Store only: Code, Notebooks, Processed datasets, Documentation, README

Repository should remain clean.

OneDrive

Store: Ministry PDFs, Official Reports, Export/Import Files, Annual Statistics, Raw Environmental Data, Large Original Files

Suggested structure:

```
      OneDrive | ├── Official_Statistic | ├── Annual_Reports | ├── Monthly_Production | ├── Export_Import | └── Fleet_Statistics | ├── Component3_Data | ├── Fisheries_Logbooks | ├── Chlorophyll_Data | ├── SST_Data | └── Rainfall_Data | └── References
```

STEP 3 - ENVIRONMENTAL DATA DECISION

Selected: Chlorophyll-a (CHL)

Reason:

- CHL is a standard indicator of ocean productivity.
- Frequently used in fisheries forecasting studies.
- Provides a single representative measure.
- Reduces feature complexity.

Not selected initially: DIATO, DINO, GREEN, HAPTO, MICRO, NANO, PICO, PROCHLO

Reason:

- These represent individual phytoplankton groups.
- More specialized variables.
- May introduce redundancy.
- Can be explored in future feature engineering.

STEP 4 - FEATURE SELECTION STRATEGY

Important principle: *Data Selection ≠ Feature Selection*

Data Selection

Choosing which datasets to collect.

Example: Logbooks, Chlorophyll, SST, Rainfall

Feature Selection

Choosing which fields enter the model.

Example: CHL, SST, Month, Catch, Effort, CPUE

STEP 5 - ANSWER FOR VIVA

How were variables selected?

Variables were selected based on fisheries research literature, domain knowledge, and relevance to tuna availability. After model development, feature importance and Explainable AI methods such as SHAP were used to evaluate the contribution of each feature.

Why not use all available variables?

Not every available variable contributes meaningful predictive information. Feature selection was performed to reduce redundancy, improve interpretability, and focus on variables with a plausible relationship to tuna availability.

Where did the fisheries data come from?

Honest answer: Fisheries logbook records were obtained through academic collaboration and subsequently validated against official fisheries statistics, species information, and ministry publications before being considered for the research.

STEP 6 - CURRENT PROJECT DECISION

At this moment:

✅ Chlorophyll-a source identified
✅ Ministry statistics organized
✅ Export/Import statistics archived
✅ Data storage strategy decided
✅ Fisheries logbook dataset identified as primary candidate
✅ Ready for fisheries column analysis

NEXT STAGE

The next task is now very clear:

```
Open each fisheries logbook year ↓ Compare column headers ↓ Find common columns ↓ Create standard schema ↓ Build fisheries master dataset
```

Version 1.1 Additions

FISHERIES DATASET DECISION

Add:

```
Current Decision:

The fisheries logbook dataset is considered the primary candidate dataset because it
contains operational fisheries records including species, quantity, weight, vessel
information, gear type, dates, and fishing location indicators.

The Ministry statistics dataset is retained for reference, validation, trend analysis and
discussion but is not currently planned as a direct model-training dataset.
```

EXPORT/IMPORT DECISION

Add:

```
Export and import datasets were reviewed but are currently not considered direct
predictor variables because they describe market behavior rather than biological or
environmental conditions affecting tuna availability.

These datasets will be retained as supporting economic context for future discussion and
reporting.
```

CHL DOWNLOAD DECISION

Add:

```
The selected Copernicus variable is:

                                    CHL
                (Mass concentration of chlorophyll a in sea water)

Other plankton group variables such as DIATO, DINO, GREEN, HAPTO, MICRO, NANO, PICO and
uncertainty variables were excluded during the initial feature selection stage to reduce
complexity and focus on the most commonly used ecological productivity indicator.
```

CURRENT RESEARCH PRIORITY

This is probably the most important update.

Add:

```
Current Priority Tasks

1. Standardize fisheries logbook datasets (2017-2024)
2. Identify common columns across years
3. Create unified fisheries schema
4. Validate tuna-related records
5. Acquire SST data
6. Acquire rainfall data
7. Construct model training dataset
```